cycle
C1
Speed Rated Intermediate Idle
Torque 100% 75% 50% 10% 100% 75% 50% 0%
Weighting
factor 0.15 0.15 0.15 0.1 0.1 0.1 0.1 0.15
3.2.6 The torque figures given in test cycle C1 are percentage values that represent for a given test mode the ratio of the required torque to the maximum possible torque at this given speed.
3.2.7 The intermediate speed for test cycle C1 shall be declared by the manufacturer, taking into account the following requirements:
.1 For engines that are designed to operate over a speed range on a full load torque curve, the intermediate speed shall be the declared maximum torque speed if it occurs between 60% and 75% of rated speed.
.2 If the declared maximum torque speed is less than 60% of rated speed, then the intermediate speed shall be 60% of the rated speed.

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SOURCE: MEPC.397(83).pdf
maximum torque speed if it occurs between 60% and 75% of rated speed.
.2 If the declared maximum torque speed is less than 60% of rated speed, then the intermediate speed shall be 60% of the rated speed.
.3 If the declared maximum torque speed is greater than 75% of the rated speed, then the intermediate speed shall be 75% of rated speed.
.4 For engines that are not designed to operate over a speed range on the full load torque curve at steady -state conditions, the intermediate speed will typically be between 60% and 70% of the maximum rated speed.
RESOLUTION MEPC.397(83) (adopted on 11 April 2025)
AMENDMENTS TO THE TECHNICAL CODE ON CONTROL OF EMISSION OF NITROGEN OXIDES FROM MARINE DIESEL ENGINES (NOx TECHNICAL CODE 2008)- 10
3.2.8 If an engine manufacturer applies for a new test cycle application on an engine already certified under a different test cycle specified in 3.2.2 to 3.2.5, then it may not be necessary for that engine to undergo the full certification process for the new application. In this case, the engine manufacturer may demonstrate compliance by recalculation, by applying the measurement results from the specific modes of the

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SOURCE: MEPC.397(83).pdf
new application. In this case, the engine manufacturer may demonstrate compliance by recalculation, by applying the measurement results from the specific modes of the first certification test to the calculation of the total weighted emissions for the new test cycle application, using the corresponding weighting factors from the new test cycle."
6 A new section 3.3 is added as follows:
"3.3 Not-to-exceed emission values within the limit area of the not -to-exceed zone
3.3.1 The boundaries, in terms of power or torque and speed, of the not-to-exceed zone at or above 25% power shall be declared to the Administration by the applicant as part of the NO x certification pack. Operation outside these not -to-exceed zone

boundaries, within the limit area of the not -to-exceed zone, shall only be permitted during starting, stopping, accelerations, deceleration, load pick -up or load rejection. However, operation below 25% power and at or above 63% speed for the E3, E2, and D2 test cycles and below 50% load for the C1 test cycle shall be permitted subject to it being shown in accordance with the requirements of 2.5 that a rational emission

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and D2 test cycles and below 50% load for the C1 test cycle shall be permitted subject to it being shown in accordance with the requirements of 2.5 that a rational emission control strategy continues to be applied.
3.3.2 The technical file as required by 2.3.4 shall additionally contain the following information:
.1 the power or torque and speed boundaries, as given by 3.3.1, within which the engine is certified to operate; and
.2 the onboard NOx verification procedure shall include means to verify that the engine only operates within the power or torque and speed boundaries as given by 3.3.1.
3.3.3 Additional to the emission testing under 3.2 the Administration may, at its discretion, require that up to three point emission values be determined at load points within the not -to-exceed zone in order to verify that the not -to-exceed zone requirements are complied with. The load points to be tested shall be agreed between the applicant and the Administration as part of the review of the NOx certification pack. Point emission values are to be determined in accordance with the procedures given

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SOURCE: MEPC.397(83).pdf
the applicant and the Administration as part of the review of the NOx certification pack. Point emission values are to be determined in accordance with the procedures given by chapter 5 and appendix X. To be acceptable each of those point emission values so determined shall not exceed the respective not-to-exceed emission limit value, , as determined using the procedure in appendix X.
Point emission value emission limit value, , at that point
3.3.4 Alternative means by which it is to be shown that a point emission value may be determined or the not -to-exceed zone requirements are complied with may be used subject to their acceptability to the Administration.
3.3.5 For member engines of engine families or engine groups first certified prior to the effective date of the requirements under this section, demonstration of compliance with the requirements of this section may be on the basis solely of documentation which is to be acceptable to the Administration."
RESOLUTION MEPC.397(83) (adopted on 11 April 2025)
AMENDMENTS TO THE TECHNICAL CODE ON CONTROL OF EMISSION
OF NITROGEN OXIDES FROM MARINE DIESEL ENGINES (NOx TECHNICAL CODE 2008)- 11 -

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SOURCE: MEPC.397(83).pdf
RESOLUTION MEPC.397(83) (adopted on 11 April 2025)
AMENDMENTS TO THE TECHNICAL CODE ON CONTROL OF EMISSION
OF NITROGEN OXIDES FROM MARINE DIESEL ENGINES (NOx TECHNICAL CODE 2008)- 11 -
Chapter 4 Approval of serially manufactured engines: Engine family and engine group concepts

7 In paragraph 4.3.8.2, sub -paragraphs 4.3.8.2.12 to 4.3.8.2.14 are added, after the existing sub-paragraph 4.3.8.2.11, as follows:
".12 multiple engine operational profiles as covered by chapter 8.
.13 base emission control strategy.
.14 auxiliary control devices."
8 Paragraph 4.3.10.5 is deleted.
Chapter 6 Procedures for demonstrating compliance with NOx emissions on board
9 In paragraph 6.2.2.3, at the end of sub-paragraph 6.2.2.3.15, the word "or" is deleted, at the end of sub -paragraph 6.2.2.3.16, "." is replaced with ",", and new sub-paragraphs 6.2.2.3.17 to 6.2.2.3.19 are added after sub-paragraph 6.2.2.3.16 as follows:
".17 list of identification references of all engine operational profiles available for the engine and, if applicable, the conditions under which each is to be used (see chapter 8 of the Code);

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SOURCE: MEPC.397(83).pdf
".17 list of identification references of all engine operational profiles available for the engine and, if applicable, the conditions under which each is to be used (see chapter 8 of the Code);
.18 list of auxiliary control devices accepted for the engine and the operating conditions under which those devices function; or
.19 the engine power or engine load and speed boundaries above 25% engine power within which the engine is certified to operate."
10 A new chapter 8 is added as follows:
"Chapter 8 Multiple engine operational profiles
8.1 Acceptance of multiple engine operational profiles
8.1.1 The switching between engine operational profiles under onboard conditions is permitted, subject to the provisions of this chapter, in the following cases:
.1 for a marine diesel engine certified to be in -service switchable between emission tiers;
.2 for a marine diesel engine certified to more than one test cycle application in accordance with 3.2 where the engine operational profile is in -service switchable based on the duty the engine is performing; or
.3 for a marine diesel engine certified to the same emission standard,

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application in accordance with 3.2 where the engine operational profile is in -service switchable based on the duty the engine is performing; or
.3 for a marine diesel engine certified to the same emission standard, the same rated power, same rated speed and the same test cycle which is in-service switchable between multiple engine operational profiles.
8.1.2 A marine diesel engine certified in accordance with 8.1.1.1 and/or 8.1.1.2 may additionally be switchable, at a particular tier and or duty, between multiple engine operational profiles in which cases the provisions of 8.1.1.3 also apply.

8.1.3 Each engine operating profile shall be identified in the technical file as

required by 2.3.4 together with the conditions, if applicable, under which each engine operating profile is to be used.

8.2 Certification of multiple engine operational profiles

8.2.1 For a marine diesel engine to which 8.1.1.1 applies, the parent engine test

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operating profile is to be used.

8.2 Certification of multiple engine operational profiles

8.2.1 For a marine diesel engine to which 8.1.1.1 applies, the parent engine test report for each tier shall be included in the technical file as required by 2.4.1.5. The parent engine specific emission value for each tier shall be entered under 1.9.6 of the Supplement to the EIAPP Certificate.

8.2.2 For a marine diesel engine to which 8.1.1.2 applies, the parent engine test report for each test cycle shall be included in the technical file as required by 2.4.1.5. The test cycles for which the engine is certified shall be shown on the EIAPP Certificate. The parent engine specific emission value for each test cycle shall be entered and identified under 1.9.6 of the Supplement to the EIAPP Certificate.

8.2.3 For a marine diesel engine to which 8.1.1.3 applies:

.1 the engine test report for each engine with the parent engine features and characteristics identified in 4.3.9 or 4.4.8, for each engine operational profile, shall be determined in accordance with